

2026 04 02
발표 자료

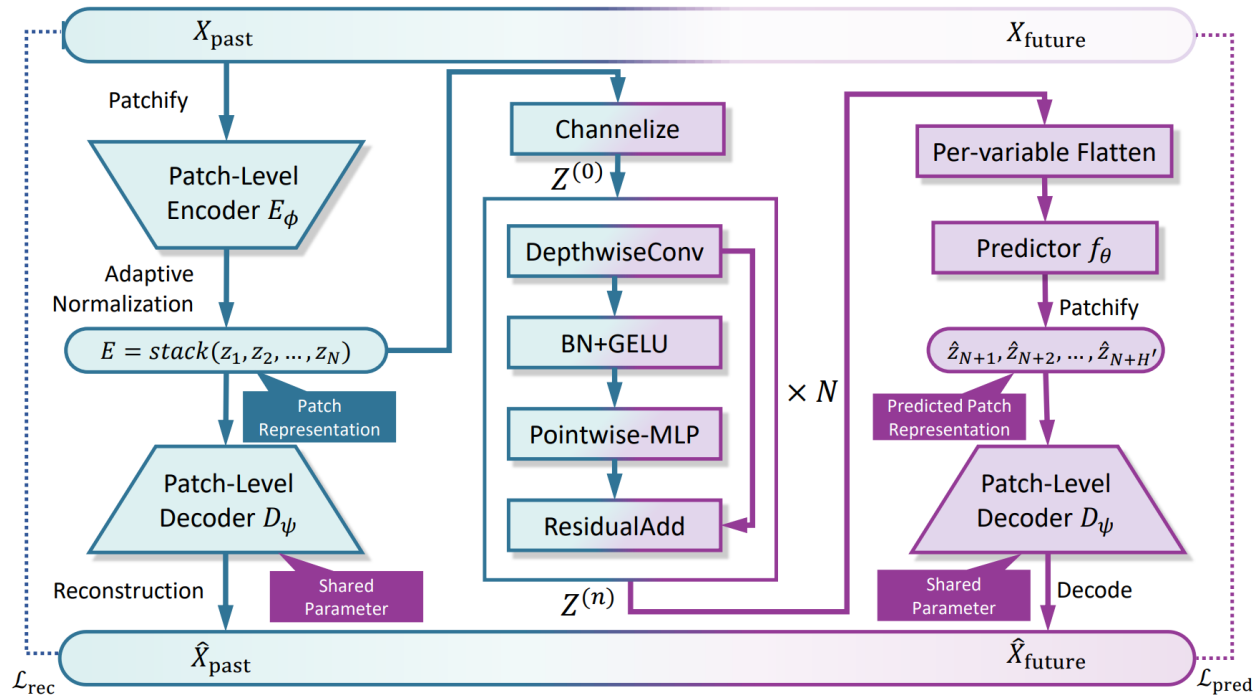
광운대학교 로봇학과
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김한서

이번 주 진행사항

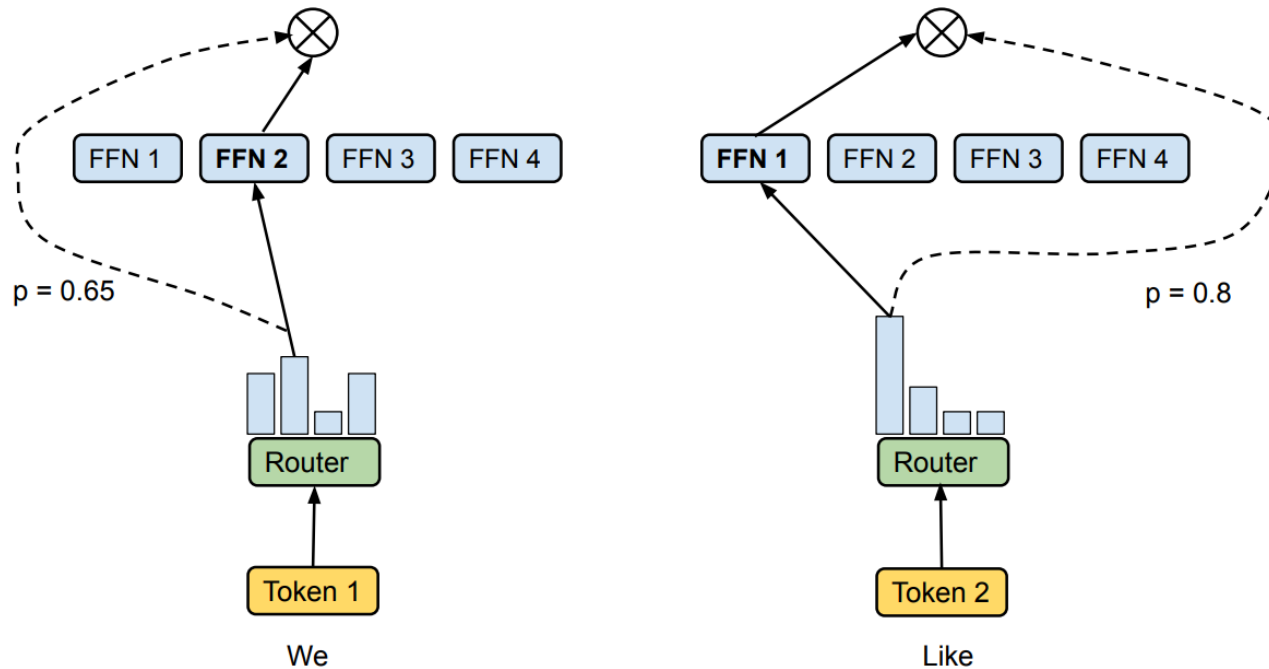
- VPNet
 - 아이디어 및 실험

Remind



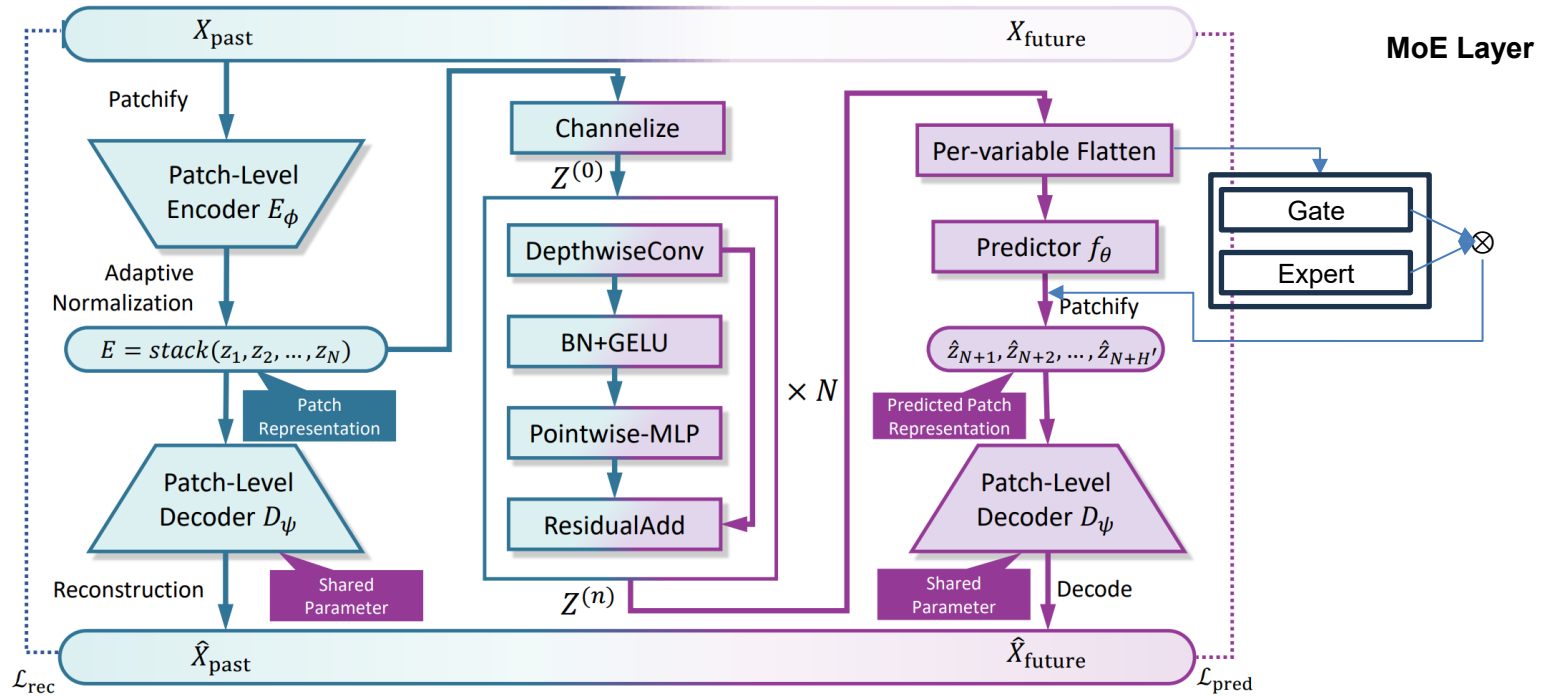
- Autoencoder, decoder를 통해 과거 데이터를 재구성하는 과정을 거치게 되고, 이 과정에서 디코더가 시계열의 데이터 분포 학습, 이후 예측 과정에서 예측값이 비정상적인 형태를 띠는 것을 방지
- Channelize 과정에서 기존 시계열 구조에서 이미지 구조로 차원 변경 후, DepthwiseConv 연산을 통해 변수 간 상관관계 추출
- Pointwise-MLP로 각 임베딩 차원의 특징값을 Mixing하여 고차원 임베딩 벡터 생성

MoE (Mixture of Experts) 설명



- 토큰을 Router에 입력하여 Router 내부의 Gate를 통과하여 각 가중치로 변환, 이후 Expert인 각 FFN에도 동일하게 토큰을 입력하여 Hidden Representation으로 변환
- 가중치와 Hidden Representation을 각각 곱한 뒤 모두 더하여 최종 벡터값 생성

아이디어



MoE Layer

- Predictor와 동일한 입력을 받아서 Gate를 통해 가중치 생성 및 Expert로 Hidden Representation 생성 각각을 곱한 최종 MoE 값과 Predictor 출력값을 더하여 사용

$$\widehat{Z}_{future} = f_\theta(X) + \alpha \cdot \Lambda \odot \sum_{i=1}^2 G_i(X) E_i(X)$$

실험 세팅

- 사용한 모델: VPNet
- 재현 실험 데이터셋: ETTh1, ETTm1, Weather

Experiment	ETTh1
Learning rate	10^{-4}
Epoch	100
Batch size	32
Loss function	MAE Loss
Seq_len	96
Pred_len	96/192/336/720
Layers	2
Patch Size	8
Hidden_len	96
Middle_len	512

VPNet 재현 실험 (ETTh1)

	ETTh1 Reproduction		ETTh1 + MoE	
Pred len	MSE	MAE	MSE	MAE
96	0.374	0.386	<u>0.375</u>	0.386
192	0.428	0.418	0.428	0.418
336	0.464	0.439	<u>0.468</u>	<u>0.440</u>
720	<u>0.471</u>	<u>0.465</u>	0.460	0.459

VPNet 재현 실험 (ETTm1)

	ETTm1 Reproduction		ETTm1 + MoE	
Pred len	MSE	MAE	MSE	MAE
96	0.311	0.341	0.311	0.341
192	<u>0.362</u>	<u>0.371</u>	0.361	0.370
336	0.386	0.390	0.386	0.390
720	0.444	<u>0.425</u>	0.444	0.424

VPNet 재현 실험 (Weather)

	Weather Reproduction		Weather + MoE	
Pred len	MSE	MAE	MSE	MAE
96	<u>0.154</u>	0.192	0.153	<u>0.193</u>
192	0.202	<u>0.240</u>	<u>0.203</u>	0.239
336	0.259	0.281	<u>0.262</u>	<u>0.284</u>
720	0.336	0.335	<u>0.339</u>	<u>0.340</u>

이후 계획

- DepthwiseConv 또는 Pointwise-MLP에 아이디어 추가

부록

주요 모델 성능 비교

Models	VPNet (Ours)		TimeKAN (2025)		TimeMixer (2024a)		iTransformer (2024a)		Pathformer (2024)		PatchTST (2023)		Crossformer (2023)		MICN (2023)		TiDE (2023)		TimesNet (2023a)		DLinear (2023)		FEDformer (2022)	
Metric	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
Weather	96	0.157 0.192	0.162	0.208	0.163	0.209	0.174	0.214	0.156 0.192	0.186	0.227	0.195	0.271	0.198	0.261	0.202	0.261	0.172	0.220	0.195	0.252	0.217	0.296	
	192	0.207 0.239	0.207	0.249	0.208	0.250	0.221	0.254	0.206 0.240	0.234	0.265	0.209	0.277	0.239	0.299	0.242	0.298	0.219	0.261	0.237	0.295	0.276	0.336	
	336	0.258 0.281	0.263	0.290	0.251	0.287	0.278	0.296	0.254 0.282	0.284	0.301	0.273	0.332	0.285	0.336	0.287	0.335	0.280	0.306	0.282	0.331	0.339	0.380	
	720	0.330 0.333	0.338	0.340	0.339	0.341	0.358	0.347	0.340 0.336	0.356	0.349	0.379	0.401	0.351	0.388	0.351	0.386	0.365	0.359	0.345	0.382	0.403	0.428	
	Avg	0.238 0.261	0.243	0.272	0.240	0.272	0.258	0.278	0.239 0.263	0.265	0.286	0.264	0.320	0.268	0.321	0.271	0.320	0.259	0.287	0.265	0.315	0.309	0.360	
Solar-Energy	96	0.177 0.178	0.234	0.290	0.189	0.259	0.203	0.237	0.202 0.225	0.265	0.323	0.232	0.302	0.257	0.325	0.312	0.399	0.373	0.358	0.290	0.378	0.286	0.341	
	192	0.196 0.206	0.274	0.309	0.222	0.283	0.233	0.261	0.235 0.245	0.288	0.332	0.371	0.410	0.278	0.354	0.339	0.416	0.397	0.376	0.320	0.398	0.291	0.337	
	336	0.216 0.219	0.299	0.324	0.231	0.292	0.248	0.273	0.272 0.272	0.301	0.339	0.495	0.515	0.298	0.375	0.368	0.430	0.420	0.380	0.353	0.415	0.354	0.416	
	720	0.228 0.225	0.295	0.318	0.223	0.285	0.249	0.275	0.255 0.256	0.295	0.336	0.526	0.542	0.299	0.379	0.370	0.425	0.420	0.381	0.357	0.413	0.380	0.437	
	Avg	0.204 0.207	0.276	0.310	0.216	0.280	0.233	0.262	0.241 0.250	0.287	0.333	0.406	0.442	0.283	0.358	0.347	0.417	0.403	0.374	0.330	0.401	0.328	0.383	
Electricity	96	0.135 0.224	0.174	0.266	0.153	0.247	0.148	0.240	0.145 0.236	0.190	0.296	0.219	0.314	0.180	0.293	0.237	0.329	0.168	0.272	0.210	0.302	0.193	0.308	
	192	0.151 0.239	0.182	0.273	0.166	0.256	0.162 0.253		0.167	0.256	0.199	0.304	0.231	0.322	0.189	0.302	0.236	0.330	0.184	0.322	0.210	0.305	0.201	0.315
	336	0.166 0.257	0.197	0.286	0.185	0.277	0.178 0.269		0.186	0.275	0.217	0.319	0.246	0.337	0.198	0.312	0.249	0.344	0.198	0.300	0.223	0.319	0.214	0.329
	720	0.196 0.284	0.236	0.320	0.225	0.310	0.225	0.317	0.231 0.309	0.258	0.352	0.280	0.363	0.217	0.330	0.284	0.373	0.220	0.320	0.258	0.350	0.246	0.355	
	Avg	0.162 0.251	0.197	0.286	0.182	0.273	0.178	0.270	0.182 0.269	0.216	0.318	0.244	0.334	0.196	0.309	0.252	0.344	0.193	0.304	0.225	0.319	0.214	0.327	
Traffic	96	0.384 0.258	0.580	0.379	0.462	0.285	0.395 0.268		0.479	0.283	0.526	0.347	0.644	0.429	0.577	0.350	0.805	0.493	0.593	0.321	0.650	0.396	0.587	0.366
	192	0.406 0.266	0.550	0.363	0.473	0.296	0.417 0.276		0.484	0.292	0.522	0.332	0.665	0.431	0.589	0.356	0.756	0.474	0.617	0.336	0.598	0.370	0.604	0.373
	336	0.429 0.275	0.559	0.363	0.498	0.296	0.433 0.283		0.503	0.299	0.517	0.334	0.674	0.420	0.594	0.358	0.762	0.477	0.629	0.336	0.605	0.373	0.621	0.383
	720	0.466 0.294	0.600	0.381	0.506	0.313	0.467 0.302		0.537	0.322	0.552	0.352	0.683	0.424	0.613	0.361	0.719	0.449	0.640	0.350	0.645	0.394	0.626	0.382
	Avg	0.421 0.273	0.572	0.372	0.485	0.298	0.428 0.282		0.501	0.299	0.529	0.341	0.667	0.426	0.593	0.356	0.761	0.473	0.620	0.336	0.625	0.383	0.610	0.376
ETTh1	96	0.374 0.386	0.374	0.397	0.375	0.400	0.386	0.405	0.390 0.390	0.460	0.447	0.423	0.448	0.426	0.446	0.479	0.464	0.384	0.402	0.397	0.412	0.395	0.424	
	192	0.428 0.418	0.416	0.422	0.429	0.421	0.441	0.436	0.437 0.419	0.477	0.429	0.471	0.474	0.454	0.464	0.525	0.492	0.436	0.429	0.446	0.441	0.469	0.470	
	336	0.464 0.439	0.451 0.443		0.484	0.458	0.487	0.458	0.497 0.445	0.546	0.496	0.570	0.546	0.493	0.487	0.565	0.515	0.491	0.469	0.489	0.467	0.530	0.499	
	720	0.471	0.465	0.463 0.463	0.498	0.482	0.503	0.491	0.494 0.461	0.544	0.517	0.653	0.621	0.526	0.526	0.594	0.558	0.521	0.500	0.513	0.510	0.598	0.544	
	Avg	0.434 0.427	0.426	0.431	0.447	0.440	0.454	0.447	0.455 0.429	0.507	0.472	0.529	0.522	0.475	0.481	0.541	0.507	0.458	0.450	0.461	0.458	0.498	0.484	
ETTh2	96	0.284 0.330	0.290	0.340	0.289	0.341	0.297	0.349	0.290 0.336	0.308	0.355	0.745	0.584	0.372	0.424	0.400	0.440	0.340	0.374	0.340	0.394	0.358	0.397	
	192	0.357 0.376	0.379	0.396	0.372	0.392	0.380	0.400	0.372 0.385	0.393	0.405	0.877	0.656	0.492	0.492	0.528	0.509	0.402	0.414	0.482	0.479	0.429	0.439	
	336	0.392 0.406	0.423	0.435	0.386 0.414		0.428	0.432	0.402	0.416	0.427	0.436	1.043	0.731	0.607	0.555	0.643	0.571	0.452	0.452	0.591	0.541	0.496	0.487
	720	0.391 0.418	0.473	0.465	0.412 0.434		0.427	0.445	0.430	0.444	0.436	0.450	1.104	0.763	0.824	0.655	0.874	0.679	0.462	0.468	0.839	0.661	0.463	0.474
	Avg	0.356 0.383	0.391	0.409	0.365 0.395		0.383	0.407	0.374 0.395	0.391	0.412	0.942	0.684	0.574	0.531	0.611	0.550	0.414	0.427	0.563	0.519	0.437	0.449	
ETTm1	96	0.313 0.342	0.329	0.366	0.320	0.357	0.334	0.368	0.318 0.349	0.352	0.374	0.404	0.426	0.365	0.387	0.364	0.387	0.338	0.375	0.346	0.374	0.379	0.419	
	192	0.362 0.371	0.363	0.380	0.361	0.381	0.390	0.393	0.365 0.372	0.374	0.387	0.450	0.451	0.403	0.408	0.398	0.404	0.374	0.387	0.382	0.391	0.426	0.441	
	336	0.385 0.389	0.390	0.404	0.390	0.404	0.426	0.420	0.401 0.397	0.421	0.414	0.532	0.515	0.436	0.431	0.428	0.425	0.410	0.411	0.415	0.415	0.445	0.459	
	720	0.444 0.424	0.460	0.443	0.454	0.441	0.491	0.459	0.460 0.432	0.462	0.449	0.666	0.589	0.489	0.462	0.487	0.461	0.478	0.450	0.473	0.451	0.543	0.490	
	Avg	0.376 0.382	0.386	0.398	0.381	0.396	0.410	0.410	0.382 0.386	0.402	0.406	0.513	0.495	0.423	0.422	0.419	0.419	0.400	0.406	0.404	0.408	0.448	0.452	
ETTm2	96	0.169 0.244	0.174	0.255	0.175	0.258	0.180	0.264	0.168 0.247	0.183	0.270	0.287	0.366	0.197	0.296	0.207	0.305	0.187	0.267	0.193	0.293	0.203	0.287	
	192	0.233 0.288	0.239	0.299	0.237	0.299	0.250	0.309	0.234 0.291	0.255	0.314	0.414	0.492	0.284	0.361	0.290	0.364	0.249	0.309	0.284	0.361	0.269	0.328	
	336	0.291 0.327	0.301	0.340	0.298	0.340	0.311	0.348	0.297 0.333	0.309	0.347	0.597	0.542	0.381	0.429	0.377	0.422	0.321	0.351	0.382	0.429	0.325	0.366	
	720	0.387 0.387	0.395	0.396	0.391	0.396	0.412	0.407	0.386 0.385	0.412	0.404	1.730	1.042	0.549	0.522	0.558	0.524	0.408	0.403	0.558	0.525	0.421	0.415	
	Avg	0.270 0.312	0.277	0.322	0.275	0.323	0.288	0.332	0.271 0.314	0.290	0.334	0.757	0.611	0.353	0.402	0.358	0.404	0.291	0.333	0.354	0.402	0.305	0.349	